

Does “Buy the Dip” Improve Dollar-Cost Averaging?

Evidence from Fair-Capital Experiments in the S&P 500, 1990–2026

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Abstract. Popular advice treats “buy the dip” as a free improvement on dollar-cost averaging (DCA): if a saver who invests every paycheck into the S&P 500 wins over a long horizon, then holding cash for a cheaper entry should win by more. That claim mixes two problems and two slogans. The first problem is a *paycheck stream* (cash arrives over time). The second is a *cash pile* already in hand. The first slogan is “do not stop buying in a crash,” which is DCA. The second is “stay underinvested until a crash,” which is timing. This paper evaluates the second slogan under a strict same-cash constraint, using daily total returns on Vanguard 500 Index Fund (VFINX) from January 1990 through August 2026, 13-week Treasury bill interest on idle cash, and 115 mechanical rules. No tradable cash-reserve overlay beats monthly DCA by an economically meaningful margin. Rules that wait for 10–20% peak drawdowns lose 3–17% of terminal wealth. A contribution-level test shows why: a dollar delayed until a 10% drawdown is invested, on average, at a 22% *higher* total-return price, not a lower one, because delay buys later in a rising market. An oracle that buys exact troughs of every $\geq 10\%$ episode still loses over the full sample. The same conclusion holds for a separate “dip sleeve” of extra capital: the fair twin is to DCA that extra, and crash sleeves capture only 66–79% of the incremental wealth. For a pile already in hand, lump-sum investment dominates both DCA of the pile and waiting for a dip. Section 8 then catalogues the genuine exceptions—lost decades, cash that would otherwise never be invested, near-continuous signals that are DCA in disguise, pull-forwards of already-committed paychecks, rebalancing of a mixed portfolio, and perfect hindsight in a bear—and shows that each is either rare, not knowable in advance, or not the strategy that retail timing advice describes.

1. Introduction

If steadily buying a broad U.S. equity index “guarantees” a long-run win, a natural corollary seems to follow: buying *more* after a decline should add profit on top of that schedule. The corollary is the content of a large popular literature on “buying the dip.” It is not a corollary of any standard model of a positive equity premium. A contribution that is delayed in order to wait for a lower price forgoes expected excess return for as long as it sits in cash. Unless mean reversion in prices is strong enough, and unless the wait is short enough, the option to delay has negative value.

The academic literature has long studied a related but different comparison: lump-sum investment of an endowment versus spreading that endowment over subsequent months (Constantinides 1979; Rozeff 1994; Brennan, Li, and Torous 2005). That is the *pile* problem. Most household savers face the *paycheck* problem. They cannot invest next year's salary today. For them, monthly DCA is not a behavioral alternative to lump sum; it is the feasible fully invested policy. "Buy the dip" in that setting is a one-sided delay. The investor does not sell a rich market—the next paycheck has not arrived—and therefore cannot run a long-short timing book. The only instrument is waiting. This paper's first contribution is to treat that constraint as the object of interest rather than as a footnote.

The second contribution is empirical design. Extra dip purchases are often backtested with extra capital, which credits timing for the return on money that would have earned the equity premium if it had been invested on schedule. Every rule here receives the same cash on the same dates. Dip extras must come from cash held back from the regular contribution, from a year-end leftover, or from an interest-free pull-forward of a future contribution that is later skipped. Signals use only prices known at the previous close; fills occur at the next close. Idle cash earns the 13-week bill. Prices are a forward total-return path starting at the actual fund NAV, not Yahoo's backward-adjusted close, so a 10% "dip" is a 10% move in a tradable total-return unit.

The third contribution is a decomposition that the popular argument gets backwards. In a positive-drift market, a 10% drawdown from a recent peak is a statement about *local* cheapness, not about the level at which a delayed paycheck is invested. Over 440 monthly contributions, a dollar that waits for a 10% peak drawdown and never flushes at year-end is filled, on average, after 263 trading days at a total-return price 22% *higher* than the price available on the contribution date. The mean relative terminal payoff of that delay is -8.8% (Newey and West 1987, $t = -2.95$). Waiting does not buy cheaper shares. It buys later shares.

The fourth contribution, in Section 8, is a disciplined account of the exceptions. The headline result—buy-the-dip as cash timing does not beat DCA—is not a claim that no path, no alternative, and no slogan ever favors buying after a decline. Earlier analysis of this sample identified a short list of cases in which dip rules help, or appear to help. That list is not a list of loopholes that restore the popular strategy. It is a taxonomy of when the comparison changes. The exceptions are: (i) a lost decade after a peak; (ii) capital that would otherwise remain in cash forever; (iii) signals that fire so often the rule is DCA; (iv) pulling forward contributions that were going to be invested anyway; (v) rebalancing a mixed portfolio that stays fully invested; (vi) an oracle in a bear market; and (vii) the slogan "keep buying the crash," which is DCA and was never in doubt. Each is taken up in detail in Section 8, with the numbers from this sample, the mechanism, and the reason it does not overturn the headline for a long-horizon S&P 500 saver.

Section 2 places the design in the literature. Section 3 states a simple delay-versus-drift inequality. Section 4 describes data and rules. Section 5 reports paycheck-stream results. Section 6 reports the pile problem. Section 7 collects robustness and limitations. Section 8 is the exceptions chapter. Section 9 concludes.

2. Related literature

Constantinides (1979) shows that a precommitted purchase schedule is dominated, for an expected-utility investor, by policies that use current information. Pye (1971) and Brennan and Solanki (1981) make related dynamic-inefficiency points. Rozeff (1994) and Williams and Bacon (1993) document that lump-sum investment of an endowment beats spreading it when the risky asset has a positive premium. Thorley (1994) labels the contrary popular argument a fallacy. The contribution of that literature is the pile problem. It does not answer whether a saver who *already* invests each paycheck should divert some of those paychecks into a crash fund.

Brennan, Li, and Torous (2005) reopen the case for DCA as a heuristic when prices mean-revert and the investor is adding to an existing portfolio of individual stocks. Their mechanism is not the one tested here. Mean reversion that helps an uninformed stock picker need not help an index investor who delays a contribution until the S&P 500 is 10 or 20% below its high. Variance-ratio estimates on this sample, reported in Section 4, show short-horizon reversal in daily returns and *continuation* at 12–36 month horizons. The waiting times of crash rules sit in the continuation region, not in a strong mean-reversion region.

Statman (1995) explains the popularity of averaging with prospect theory, regret, and self-control rather than with a claim of higher expected wealth. Dubil (2005) emphasizes risk reduction over cost savings. Milevsky and Posner (2003) map averaging into an Asian-style claim, which can look attractive conditional on a terminal price—an ex post comparison that is not a feasible policy. None of these papers evaluates mechanical drawdown, moving-average, or RSI overlays on a contribution stream with a same-cash constraint.

A separate literature studies dollar-weighted versus time-weighted returns (Dichev 2007; Friesen and Sapp 2007; Frazzini and Lamont 2008). Aggregate investors earn less than buy-and-hold because they add capital after high returns. “Buy the dip” is an attempt to reverse that gap by adding capital after low returns. The attempt still requires the capital to have been withheld beforehand. Dichev’s gap is about the timing of flows that actually occurred. This paper is about a counterfactual flow policy for a household that can choose when to deploy a given paycheck.

Return predictability and tactical allocation are also adjacent rather than identical. Campbell and Shiller (1988), Fama and French (1988), and Poterba and Summers (1988) document slow-moving components in prices. Goyal and Welch (2008) show that many predictive regressions fail out of sample. Faber (2007) and the time-series momentum literature (Moskowitz, Ooi, and Pedersen 2012; Hurst, Ooi, and Pedersen 2017) study fully invested portfolios that switch between an index and cash or that overlay trend. Those strategies *sell* as well as buy. Contribution-timing cannot sell the peak. Moreira and Muir (2017) volatility-managed portfolios similarly reallocate existing wealth. Section 8 returns to rebalancing as an exception precisely because it is this other mechanism.

Behavioral work on myopic loss aversion (Benartzi and Thaler 1995) and on long-horizon investing when returns are predictable (Barberis 2000) explains why households might want a rule that keeps them in the market. That is an argument for DCA during a crash, not for a cash pile earmarked for one.

3. Theoretical framework

Let the risky asset have price P_t with expected excess return $\mu > 0$ over cash, which earns r_f . A contribution of one dollar invested immediately at date t is worth P_T/P_t at a common horizon T . The same dollar delayed until a stopping time $t+\tau$, earning cash in the interim, is worth

$$\exp(r_f \tau) \times P_T / P_{t+\tau} \quad (1)$$

The relative terminal payoff of delay versus immediate investment is therefore

$$\Delta = \exp(r_f \tau) \times (P_t / P_{t+\tau}) - 1. \quad (2)$$

If, over the wait, the total-return price has expected drift $r_f + \mu$, then $E[P_{t+\tau}/P_t] \approx \exp((r_f + \mu)\tau)$ and

$$E[\Delta] \approx \exp(-\mu \tau) - 1 < 0. \quad (3)$$

Delay is then strictly dominated in expectation. Mean reversion can overturn (3) only if it compresses $E[P_{t+\tau}/P_t]$ enough to offset the forfeited premium. A 10% “discount” that arrives after the market has already risen 30% is not a discount in the units of equation (2). It is a local drawdown at a higher level.

Two comparative statics follow. First, $E[\Delta]$ is decreasing in expected wait τ , hence in the rarity of the trigger. Crash rules (20–30% peak drawdowns) should do worse than noisy high-frequency rules (three consecutive down days), not better. Second, $E[\Delta]$ is decreasing in μ . On an asset with a large premium, waiting is expensive. On cash itself, waiting is free. That is why “buy the dip versus T-bills” and “buy the dip versus DCA” are different hypotheses. Section 8 treats them as different exceptions.

A portfolio that holds back a fraction α of each contribution has, to first order, a terminal gap of about $\alpha E[\Delta]$ relative to full DCA, plus a covariance between the size of the reserved book and subsequent returns. If reserved cash is largest after long bull markets—exactly when crash rules have not fired—the covariance is unfavorable. That is the lost-decade converse: reserved cash is useful when the subsequent path is a long sideways or down market. That path is one of the exceptions, not the unconditional mean.

4. Data and empirical design

4.1 Prices

The investment vehicle is VFINX, the Vanguard 500 Index Investor share class, daily from 2 January 1980 (warmup) with live contributions from 2 January 1990 through 14 August 2026 (9,219 trading days, 440 months). Yahoo Finance supplies the fund’s unadjusted NAV, the backward-adjusted close, the S&P 500 price index, and the 13-week Treasury bill discount. The backward-adjusted close is not used as a trade price. It reprints 1980 as \$4.41 so that the last day equals today’s NAV (\$719.22). Mixing that series with raw NAV makes “average cost” and “10% dip” incomparable. The backtest uses a forward total-return price: start at the actual NAV, then compound each day’s adjusted-close return so distributions remain in the path. Live-window total-return price runs from \$54.18 to \$2,339.41. Every strategy detects dips on that series and fills on that series, except a robustness set that takes the signal from the

S&P 500 price index and still fills on the total-return price. Dip extras fill at the next close. Monthly DCA fills at the month-start close. Fractional shares, no fees (immaterial for monthly VFINX in this period). Idle cash compounds at the bill rate.

4.2 The asset

Over the live window the total-return CAGR is 10.83% with daily-annualized volatility 17.99%, Sharpe ratio 0.52 relative to bills, and a maximum drawdown of −55.3%. Mean bill yield is 3.80%, so the sample equity premium is 7.03% per year. There are 42 peak-to-recovery episodes of at least 5%, 14 of at least 10%, four of at least 20%, and three of at least 30%. Conditional on a $\geq 10\%$ episode, mean trough depth is −21.8% and mean days to trough is 111. Overlapping variance ratios in the sense of Lo and MacKinlay (1988) are below one at daily and weekly horizons (short-run reversal) and above one at 12–36 month horizons (1.07, 1.18, 1.17). Crash waits of several hundred trading days therefore sit where continuation, not reversal, is the sample feature. Table 1 collects these moments.

Table 1. Sample moments, VFINX total return, January 1990–August 2026

Moment	Value
Live window	2 Jan 1990 – 14 Aug 2026
Trading days / months / years	9,219 / 440 / 36.61
CAGR	10.83%
Volatility (daily, annualized)	17.99%
Mean 13-week bill	3.80%
Equity premium vs bills	7.03%
Sharpe (excess vs bills)	0.52
Maximum drawdown	−55.3%
Episodes $\geq 5\%$ / 10% / 20% / 30%	42 / 14 / 4 / 3
Variance ratio, monthly $q=12$ / 24 / 36	1.07 / 1.18 / 1.17

Notes: Total-return price is NAV compounded at adjusted-close returns. Episodes are peak-to-recovery windows on that series. Variance ratios are overlapping Lo and MacKinlay (1988) point estimates.

4.3 Strategy families

The catalog has 115 rules in seven families. *DCA* invests 100% of each \$1,000 on the first trading day of the month (\$440,000 contributed). *Powder* invests 50–80% on schedule and holds 20, 30, or 50% for a signal (peak drawdown 5–30%, 252-day high, 50- or 200-day moving average, Wilder RSI(14) below 30 or 25, one-day $\leq -2\%$, five-day $\leq -5\%$, 21-day $\leq -10\%$, three consecutive down days, two-sigma down day, and 10/20/30% tranches). Some powder rules flush leftovers at year-end. *Wait / only-dip* holds 100% until the signal, with or without a year-end flush. *Pull-forward* remains fully invested on schedule but, on the rising edge of a new dip episode, invests the next one or three months now and skips those later months—one advance at a time, repaid from subsequent paychecks, no standing 0% loan. *Sleeve* keeps the original \$1,000 in DCA and adds \$200, \$300, or \$500 that only buys dips; the fair

twin is DCA of the combined cash. *Oracle* deploys reserved powder on the exact trough of every $\geq 10\%$ peak-to-trough episode, using future information, as an upper bound rather than a tradable rule. A robustness set repeats several rules with newspaper S&P 500 index drawdowns as the signal.

4.4 Inference

Portfolio terminal wealth is the household object, but 36 years is one overlapping path. The paper therefore also treats each of the 440 monthly contributions as a unit. For each month-start date t and each signal, the first subsequent fill date is recorded (signal, year-end flush, or sample end). The relative terminal value of delay versus immediate investment is Δ in equation (2). Means, medians, win rates, and Newey and West (1987) t -statistics with 12 monthly lags are reported. Rolling ten-year windows (26 overlapping starts) and calendar decades are reported as path variation, not as independent trials. With 115 portfolio rules, a Bonferroni cutoff of 0.05/115 would not treat a +0.16% terminal gap as a discovery. The contribution-level tests of 10–20% delays reject in the *negative* direction.

5. The paycheck problem: main results

DCA of \$1,000 a month finishes at \$5.105 million (internal rate of return 10.96%, 2,182 shares, average cost \$201.64 in total-return units). Figure 1 plots selected paths relative to that baseline. Figure 2 plots terminal wealth. Figure 3 is a heatmap of reserve $\times$ peak-drawdown trigger. The pattern matches equation (3): larger reserves and rarer triggers finish with less wealth, with the qualification in Figure 3 that a -30% trigger is slightly less damaging than -20% because it almost never fires. A 50% reserve for 20% crashes finishes at \$4.227 million (-17.2%). A 30% reserve for 10% drawdowns finishes at \$4.791 million (-6.2%). Waiting for a 10% dip and flushing at year-end finishes at \$4.956 million (-2.9%). The best cash-reserve overlay is a 50% reserve for three consecutive down days: +0.08%. Among all same-budget tradable rules, the best is only-dip on three down days, never flushing: +0.16%. Pull-forward of three months at a 10% drawdown is +0.13%. These are rounding error. They are Exceptions 3 and 4 in Section 8, not evidence that crash timing works.

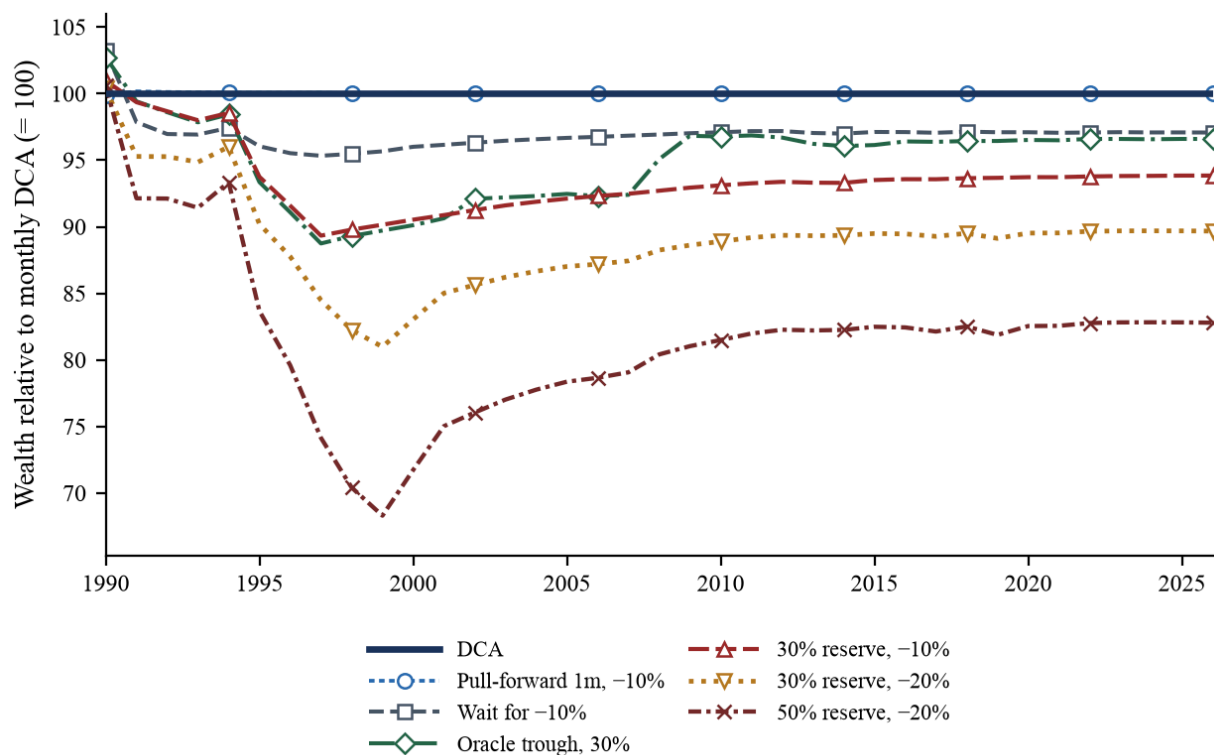


Figure 1. Wealth relative to monthly DCA (=100), selected rules, 1990–2026. Colors are paired with distinct line styles and markers. The vertical axis is unclipped: the 50% reserve for 20% crashes falls to about 68 in the late 1990s. Pull-forward tracks the DCA line (open circles). Crash reserves gap in the 1990s bull market and never close. The oracle recovers in 2000–09 and still ends below 100.

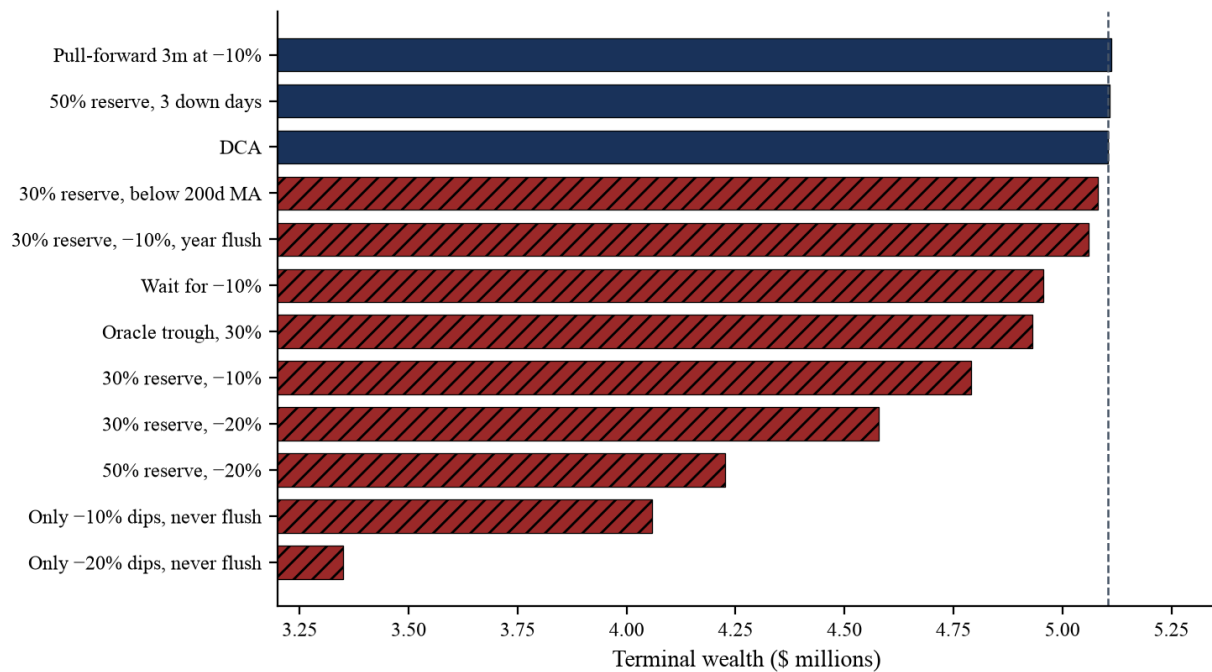


Figure 2. Terminal wealth. The dashed line is DCA at \$5.105 million. Blue bars are at or above DCA; red bars are below. Only near-continuous or pull-forward rules sit on the right of the line, and then by 0.1%.

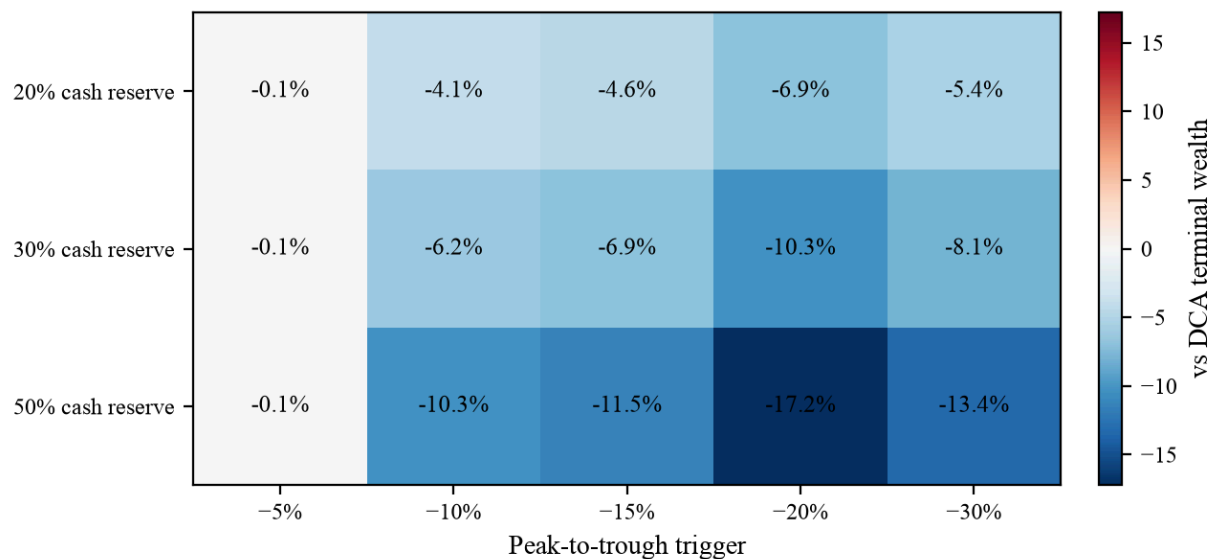


Figure 3. Percent versus DCA terminal wealth for powder rules that deploy on peak drawdowns. Every cell is negative. The -5% trigger is near zero because it is almost always invested. The penalty peaks at -20%, then eases slightly at -30% because that trigger is rare—not because deep-crash timing is a better policy over the full sample (see Section 8.1).

Table 2 reports the contribution-level test. This is the clean treatment effect of delay. Waiting for a 10% drawdown without a year-end flush has mean $\Delta = -8.84\%$ ($t = -2.95$, $p = 0.003$), mean wait 263 days, and a mean fill price 22.1% *above* the contribution-date price. A 20% rule: mean $\Delta = -16.21\%$ ($t = -3.61$), wait 789 days, fill price +57.7%. Year-end flushes cut the damage by capping τ , but do not reverse the sign. Three down days: mean $\Delta = +0.04\%$ ($t = 0.18$), wait 14 days. The oracle that buys the

next true trough: mean $\Delta = +4.69\%$ with $t = 0.68$ —positive on average, indistinguishable from noise, and not a tradable rule. Figure 4 plots these means with 95% Newey and West (1987) bands.

Table 2. Contribution-level delay versus immediate investment (440 months)

Rule	Mean Δ	NW t	p	Win %	Mean wait (days)	Mean ΔP while waiting
Three down days, never flush	+0.04%	+0.18	0.86	49.3	14	+0.2%
5% drawdown, year flush	−0.41%	−0.89	0.38	28.4	34	+1.0%
Below 200-day MA, year flush	−1.49%	−2.19	0.03	37.7	67	+2.8%
RSI(14)<30, year flush	−1.89%	−2.03	0.04	43.4	88	+3.8%
10% drawdown, year flush	−2.35%	−3.04	0.002	24.5	73	+3.8%
20% drawdown, year flush	−3.41%	−3.42	0.001	22.7	104	+5.5%
10% drawdown, never flush	−8.84%	−2.95	0.003	28.6	263	+22.1%
30% drawdown, never flush	−10.88%	−2.00	0.05	36.4	1,024	+52.7%
20% drawdown, never flush	−16.21%	−3.61	<0.001	30.2	789	+57.7%
Oracle trough (lookahead)	+4.69%	+0.68	0.50	50.7	577	+22.1%

Notes: Δ is relative terminal value of \$1 delayed until the rule’s fill versus \$1 invested at the month-start close, using the same terminal total-return price. Newey and West (1987) standard errors use 12 lags. “Mean ΔP while waiting” is the average of $P_{\text{fill}}/P_{\text{contribution}} - 1$. A positive number means the delayed dollar bought at a higher total-return price. Win % is the share of months with $\Delta > 0$.

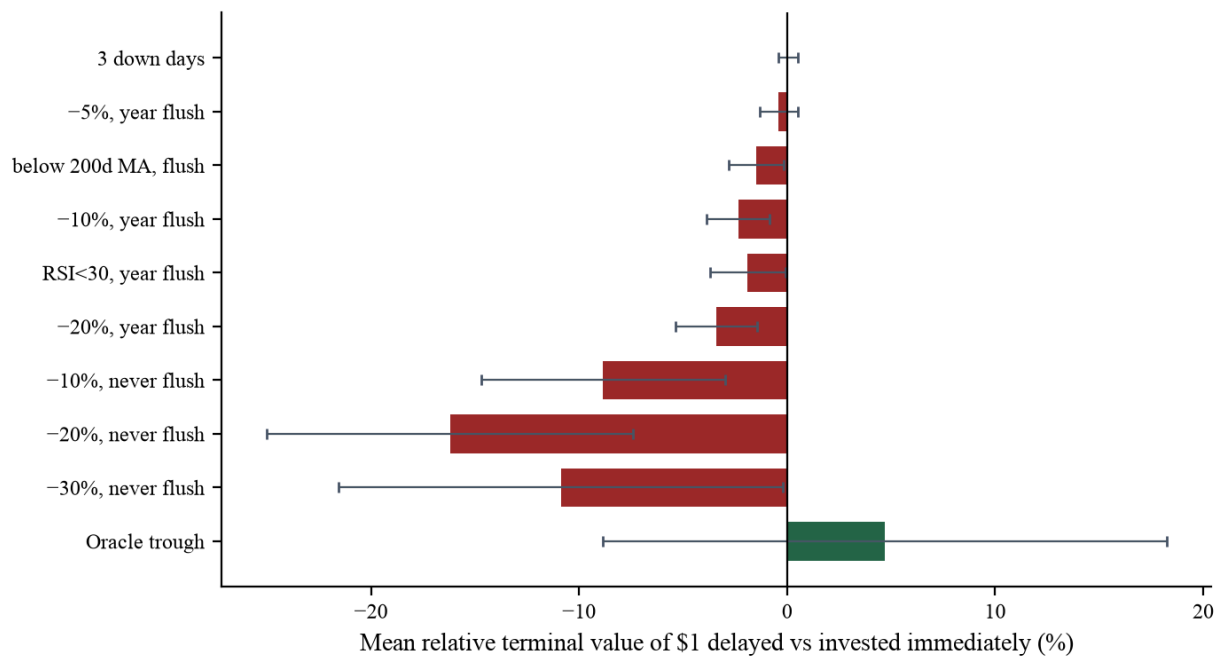


Figure 4. Mean relative terminal value of delaying a \$1 contribution, with 95% Newey and West (1987) bands. Tradable crash rules lie to the left of zero. The oracle lies to the right with a band that includes large losses.

Figure 5 (cost versus cash) is the portfolio counterpart of Table 2. Rules that hold more cash have *higher* average purchase costs than DCA, not lower. Even the oracle sits at a higher average cost. In a rising to-

tal-return series, later is not cheaper. Figure 6 shows the sleeve experiment. Incremental wealth from an extra \$300 a month that waits for dips, as a percent of the wealth obtained by DCAing that extra \$300: 100% for three down days, 98% below the 200-day average, 97% for a 10% dip with year-end flush, 79% for a 10% peak rule that never flushes, 66% for a 20% peak rule. Extra capital does not rescue timing. It makes the opportunity cost of delay visible as a capture ratio below 100.

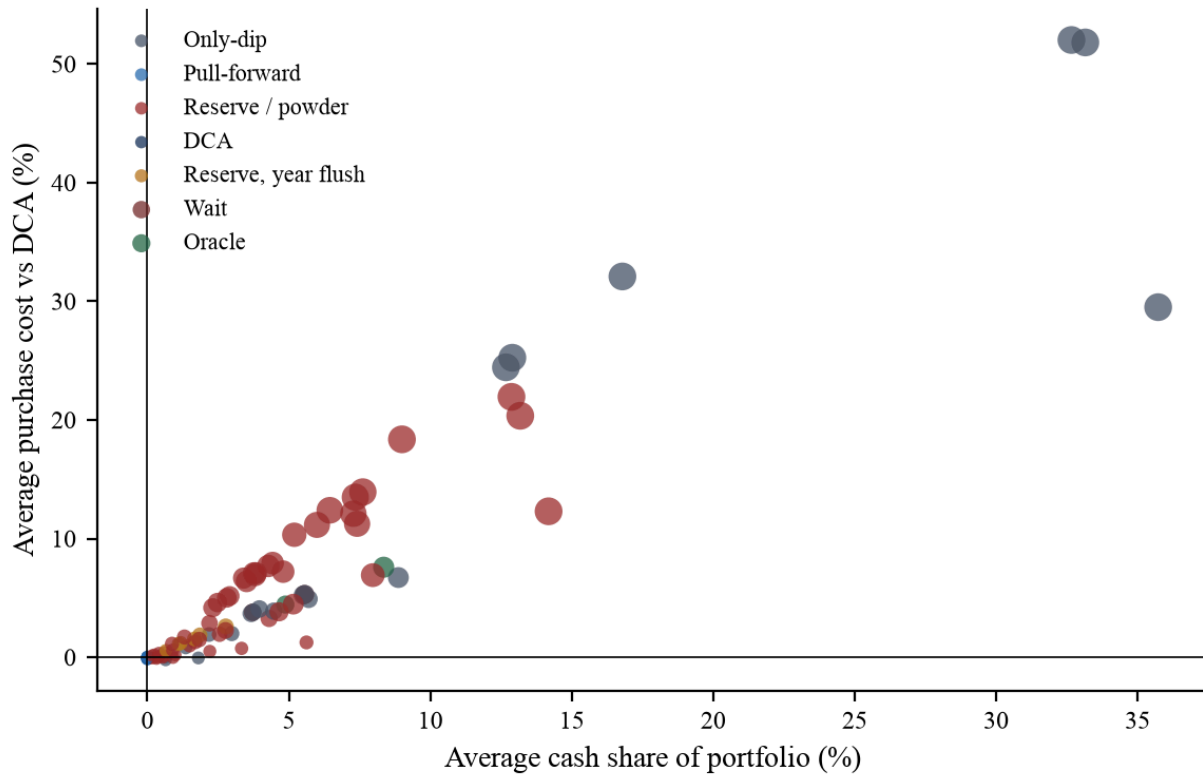


Figure 5. Average purchase cost versus DCA against average cash share of wealth. Marker size scales with $|\text{terminal gap}|$. The popular story (more cash, lower average cost) is the opposite of the scatter.

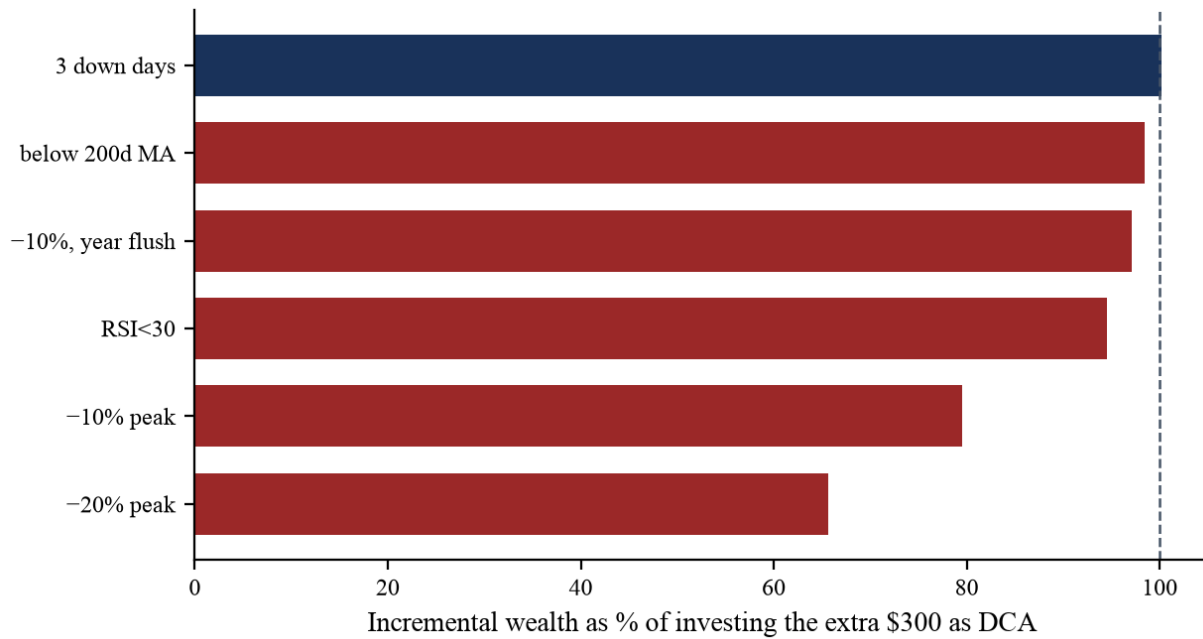


Figure 6. Incremental terminal wealth of a \$300 monthly dip sleeve as a percent of investing that \$300 as DCA. The dashed line is 100%. Deeper triggers capture less of the extra premium.

Rolling ten-year windows (Figure 7) confirm that crash rules beat DCA only in a minority of starts, with negative medians. Pull-forward of three months beats DCA in 88.5% of windows, with a median of +0.08%—a high win rate on an economically empty margin, discussed as Exception 4. Only-dip at 30% with no flush “wins” half the windows, with median −1.1% and mean −5.8%: a right tail from lost-decade starts and a left tail from bulls, which is Exception 1 rather than a usable rule.

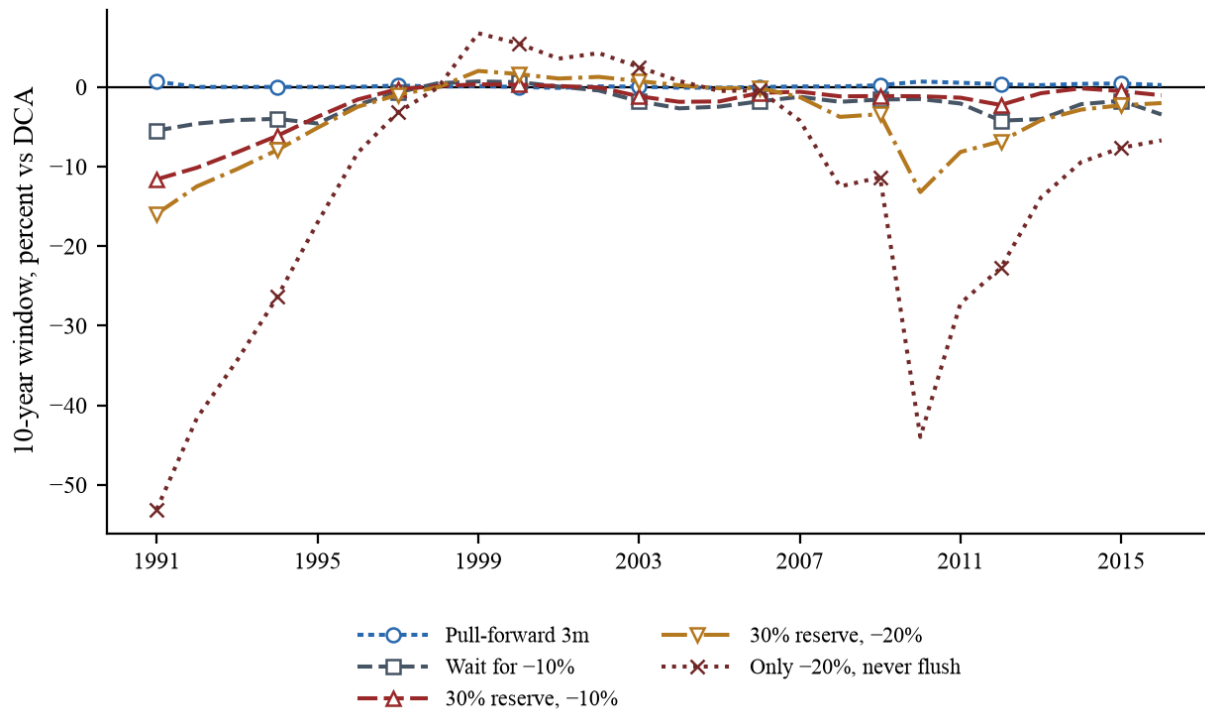


Figure 7. Overlapping ten-year windows, percent versus DCA. Crash reserves and only-dip at -20% are below zero except around starts that feed the 2000s. Pull-forward hugs zero.

6. The pile problem

A household that already holds the cash faces the opposite comparison. Spreading a pile is taking equity risk later (Rozeff 1994). Waiting for a dip is taking it even later, and only if a trigger appears. Table 3 reports 380 five-year and 320 ten-year windows, \$100,000 at each month-start, bills on idle cash, same total-return series.

Table 3. Cash pile of \$100,000: percent versus lump sum

Policy	5y win vs lump	5y median	5y mean	10y win vs lump	10y median	10y mean
Lump sum (day 1)	—	0	0	—	0	0
DCA over 12 months	23.4%	-4.4%	-3.0%	24.1%	-4.6%	-2.7%
DCA over 24 months	17.4%	-8.3%	-5.8%	18.8%	-8.6%	-5.5%
Wait for -10%	34.5%	-8.5%	-12.6%	35.3%	-10.5%	-14.9%
Wait for -20%	34.5%	-14.4%	-15.9%	27.2%	-19.8%	-22.3%
Wait -10% , else buy at 12 months	35.5%	-6.0%	-5.2%	36.6%	-6.0%	-5.0%
Wait -20% , else buy at 12 months	25.5%	-9.1%	-5.3%	24.7%	-9.1%	-5.6%

Notes: Windows begin on month-starts from January 1990. Five-year last start August 2021 ($n=380$); ten-year last start August 2016 ($n=320$). Win is the share of starts with higher terminal wealth than lump sum. Wait rules buy the entire remaining pile at the next close after the post-decision peak is down 10 or 20%, with no cap unless stated.

Waiting for a 10% dip beats lump sum more often than 12-month DCA does (35% versus 24% of ten-year starts), because a subset of starts is followed by a crash. The left tail is worse: the 10th percentile versus lump is −51% with unlimited wait for −10%, versus −11% for 12-month DCA. Median and mean still favor lump sum. Capping the wait at 12 months trims the disaster and still loses 6–9% at the median. For a pile, buy-the-dip is not an exception that rescues the investor. It is the worst of the three simple policies on typical outcomes. The exception that people want—waiting is safer and still richer—does not appear in this table. What does appear is a sleep-insurance interpretation of 12-month DCA: historically about five percent of ten-year terminal wealth.

7. Robustness

Newspaper drawdowns on the S&P 500 price index as the signal, with fills still on VFINX total return, produce the same ranking: 30% powder at a 10% index drawdown, −6.1%; wait for a 10% index dip with year-end flush, −2.9%. Year-end flushes dominate never-flushing for rare triggers because they cap τ . Tranching 10/20/30% does not help (30% powder tranches, −10.6%). Unit and one-third deployment rules that dribble reserved cash into a drawdown still lose. Next-day execution is conservative relative to a same-close fill; relaxing it would not reverse crash-rule gaps of several percent.

7.1 Limitations

The results are a historical record, not a forecast. Nothing in the design guarantees that a ranking observed from 1990 through 2026 will hold for the next 36 years. If the U.S. equity premium over bills were to collapse, or if a lost decade like 2000–2009 were the typical path rather than the exception, cash-timing would look better and lump-sum investment of a pile would look worse. The paper’s mechanism—delay forfeits expected excess return unless mean reversion more than pays for the wait—survives that caveat only insofar as a positive premium remains. The sample cannot speak to a world in which that premium is zero or negative for a long time.

Other limits are more mundane but material. The live window is one fund, one country, and one overlapping 36-year path; decade splits (Figure 8) and rolling ten-year windows describe variation inside that path, not independent draws from the same distribution. Yahoo Finance total returns are a convenient proxy, not the CRSP value-weighted market. The tests ignore taxes, bid–ask costs, and the possibility that a household cannot hold through a 50% drawdown. Mechanical price rules are not a test of discretionary valuation, security selection, or every untried filter. Section 8 catalogues when dip-waiting helped *inside* this sample; it is not a license to treat those cells as a trading plan for an unseen future.

8. When buying the dip can help: a taxonomy of exceptions

The preceding sections answer a specific question: relative to investing each S&P 500 contribution as soon as it arrives, does a mechanical rule that *holds cash for a cheaper print* raise terminal wealth? The answer is no, unconditionally, in this sample. That is not the same as the claim that no investor, on no path, with no alternative, ever benefits from buying after a decline. Popular discussion collapses those claims. This section separates them. The list is the set of exceptions identified in the design of this

project. Each subsection states the comparison that must be true for the exception to operate, the numbers, the mechanism, and the reason the exception does not restore “keep dry powder for the S&P 500 crash” as a default rule.

A useful filter, used throughout, is: *helpful relative to what?* Buy-the-dip can beat T-bills and still lose to DCA. It can beat DCA on 2000–2009 and still lose on 1990–2026. It can match DCA when it is DCA. Those are different sentences.

8.1 Exception 1: a lost decade after a peak

This is the only calendar decade in the live sample in which crash-waiting raised terminal wealth relative to DCA (Figure 8). From January 2000 through December 2009, a saver who invested \$1,000 a month on the first trading day finished at \$127,691. The same saver who held every contribution in bills until a 20% peak drawdown, and never flushed, finished at \$134,681 (+5.47%). Waiting for 30% finished at \$151,766 (+18.85%). A 50% reserve deployed only on 20% crashes finished at \$131,186 (+2.74%). A \$300 sleeve that waited for 20% drawdowns beat its same-budget DCA twin by +1.26%. The oracle that bought exact troughs of $\geq 10\%$ episodes with 30% powder finished +15.66%.

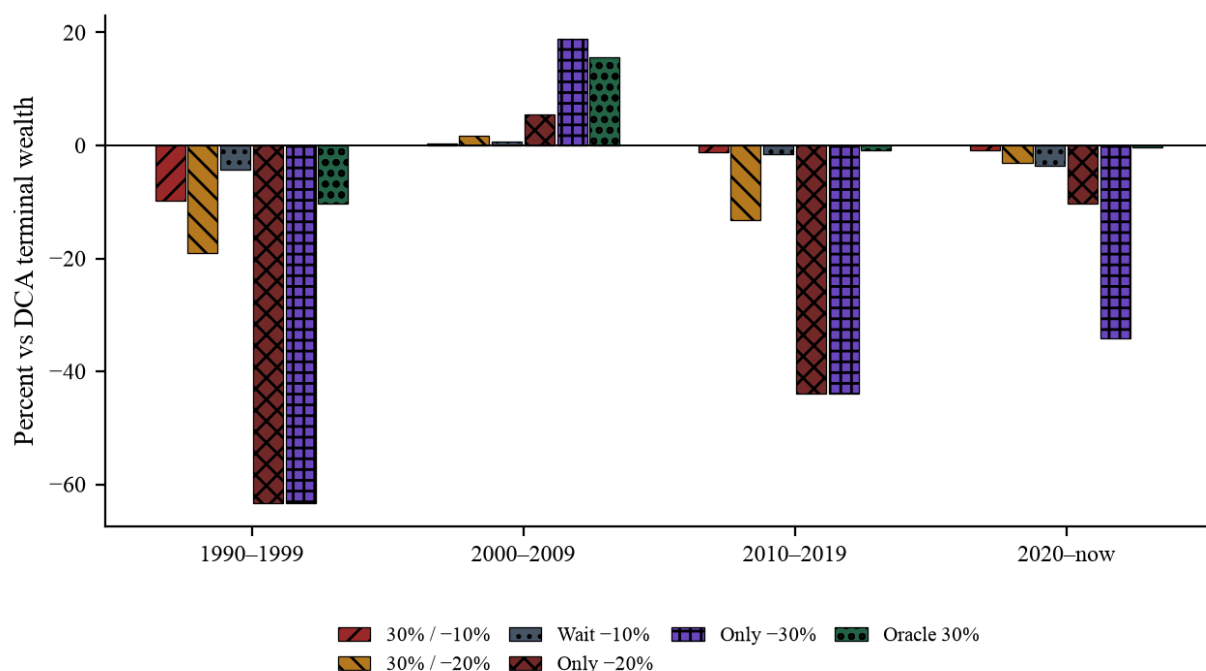


Figure 8. Calendar decades, percent versus DCA. The zero line is DCA (omitted as a bar). Colors and hatch patterns identify each series. The only decade in which crash-waiting helps is 2000–2009. That cell is Exception 1, not the full-sample policy.

The mechanism is the converse of equation (3). In that decade the realized premium on being invested early was low, then violently negative, then a rebound from a deep trough. Cash drag was cheap because the thing you were not holding did not grind up. The 2000–02 and 2007–09 declines were large enough that a 20% and 30% trigger actually fired, and the subsequent recovery was large enough that shares

bought near those troughs compounded. This is the path on which “I wish I had dry powder” is an ex post true statement.

It is not a rule that could have been followed only in that decade without a forecast of that decade. The same only-dip 20% policy, run as an isolated strategy on the other decades, finished −63.4% versus DCA in 1990–1999, −44.0% in 2010–2019, and −10.3% in 2020–August 2026. The 30% only-dip rule is identical to the 20% rule in the two long bulls (neither deep crash arrived in time) and then −34.2% in 2020–now, when a 30% episode was late relative to the 2020 rebound the DCA investor had already bought. Over the full 1990–2026 path the 20% never-flush rule is −34.4% versus DCA; the 30% rule is −26.9%. The lost decade is real. It is also one decade in four, and it is not labeled in real time as a lost decade until it is over. Rolling ten-year windows that *start* near the 2000 peak inherit some of this benefit; windows that start in the mid-1990s or after 2009 do not.

Two further caveats belong with this exception. First, even in 2000–2009 the *paycheck* investor who simply kept DCAing still compounded contributions through the middle of the crash, which is the “keep buying” slogan of Exception 7. Crash-waiting beat that schedule in that decade; it did not require a forecast only of “there will be a dip,” but of “the next ten years will not pay you for being invested early.” Second, the sleeve result in 2000–2009 is +1.26% versus putting the extra \$300 into DCA—an order of magnitude smaller than the only-dip 30% result, because most of the money was still being invested on schedule. Dry powder large enough to matter in a lost decade is dry powder large enough to devastate a bull decade (50% reserve, 20% trigger: −31.7% in the 1990s, −22.0% in the 2010s).

8.2 Exception 2: the money would otherwise stay in cash

If the alternative is not DCA but *never buying equities*, then almost any dip rule that eventually buys the index beats T-bills over this sample. A saver who put each \$1,000 into bills and never bought VFINX finished at \$738,160. DCA finished at \$5.105 million, about 6.9 times larger. The “failed” only-dip 20% policy that never flushes—the rule that lost 34% to DCA—still finished at \$3.348 million, about 4.5 times the all-cash terminal. Only-dip at 10% never-flush finished at \$4.058 million. In that comparison, buying the dip “worked.” The comparison is not the one the popular corollary states, and it is not the one a saver who already intends to own the S&P 500 faces.

The sleeve experiment is the right test for “I have extra money that I will only deploy on a crash.” Relative to leaving that extra in bills forever, the sleeve is much better. Relative to DCAing the extra, a 10% peak sleeve keeps 79% of the incremental wealth and a 20% peak sleeve keeps 66%. So the exception has a nested structure:

- Dip-buying extra cash » leaving extra cash in T-bills for 36 years.
- Dip-buying extra cash < DCAing extra cash, except for near-continuous signals (Exception 3) and the 2000–09 subsample (Exception 1).

This is also why advice aimed at a person who is scared and 100% in cash can be useful without contradicting the paper. Telling that person to buy a 20% drawdown may be what gets them into the market at all. The welfare gain is “entered equities” versus “never entered,” not “timed the index” versus “bought

every month.” Once the person would have been a DCA saver anyway, the exception expires. The capture ratios above are the cost of refusing to let the exception expire.

8.3 Exception 3: the signal fires so often that the rule is DCA

Several rules in the catalog finish within about two tenths of a percent of DCA: only-dip on three consecutive down days (+0.16% never flush, +0.13% year flush); 20–50% powder on three down days (+0.03 to +0.08%); 5% peak-drawdown powder (about −0.05 to −0.13%). Contribution-level Δ for three down days is +0.04% with $t = 0.18$ and a 14-day mean wait. That is not a working timing model. It is DCA with a short, noisy delay that sometimes buys a one-day bounce and sometimes buys a continued slide, netting noise.

The economic content of this exception is a warning about backtest mining. If a researcher searches 115 rules, some high-frequency filters will finish a few basis points above the baseline. They will also look, in a table, like “buy-the-dip works.” They work because they do not wait for a dip in any ordinary-language sense. Three down days occurred on 332 deployment days in the powder tests; a 20% peak rule deployed on 80. Equation (3) says the first should look like DCA and the second should look like delay. That is what the catalog shows. Treating Exception 3 as support for crash timing is a category error.

8.4 Exception 4: pulling forward money that was going to be invested anyway

Pull-forward is the one family that is fully invested on the calendar and, on the first day of a new dip episode, advances the next one or three months of contributions, later skipping those months. There is no extra outside capital and no standing loan: one advance at a time, repaid from subsequent paychecks. Over the full sample, a one-month pull-forward at 10% drawdown finishes +0.03%; a three-month pull-forward at 10% finishes +0.13%; a one-month pull-forward on the 200-day average finishes +0.15%. In rolling ten-year windows the three-month 10% rule beats DCA in 88.5% of starts, with median +0.08% and 90th percentile +0.50%.

This is a genuine, small, same-cash improvement in this sample. It is also not “holding dry powder.” The investor never builds a crash fund. The investor slightly changes the *date* of contributions that were already committed. The mechanism is a few weeks of extra market exposure around the start of drawdowns, financed by shifting exposure away from later months in the same episode. Because the S&P 500 has a positive premium, a small, bounded, interest-free shift of cash that would have been invested shortly anyway can have a positive mean. The magnitude is a few tenths of a percent of terminal wealth over 36 years. It does not accumulate into a reason to sit in bills waiting for −20%. If an institution already auto-invests and can lawfully accelerate the next one to three scheduled lots when a drawdown begins, this exception is the closest thing in the catalog to a free lunch, and the lunch is a rounding error. If the household’s alternative is to hold 30% of each paycheck in cash for years, Exception 4 is not that policy.

8.5 Exception 5: rebalancing a mixed portfolio that stays fully invested

A 60/40 investor who rebalances after equities fall is “buying the dip” in English and is doing something the tests in this paper never allowed. Rebalancing sells what rose and buys what fell *with wealth that is*

already in the market. The book remains fully invested (or fully invested in a chosen mix). There is no bill drag on a reserved contribution stream. The mechanism is diversification plus a mechanical contrarian flow between two assets that both have risk premia, not a delay of equity contributions into cash.

This paper does not backtest 60/40, risk-parity, or volatility targeting, and it does not claim those strategies lose. It claims they are not identified by a VFINX contribution-timing horse race. Faber (2007) moving-average timing and time-series momentum similarly sell the index for bills or bonds; they are tactical asset allocation of existing wealth. They have mixed out-of-sample records and they require the investor to be out of equities in long bulls—the same cost that crushed crash-waiting in 1990–99 and 2010–19—but they are a different object. Anyone who cites this paper as showing that “rebalancing does not work” is misreading Exception 5. Anyone who cites rebalancing as showing that “dry powder for the S&P works” is smuggling in a fully invested mix and calling it cash timing.

8.6 Exception 6: perfect hindsight in a bear

The oracle buys the exact trough of every $\geq 10\%$ peak-to-trough episode. It uses future information: the trough is known only after the series makes a new high. It is an upper bound on dip-timing skill for a reserved-cash policy, not a strategy. In 2000–2009, 30% oracle powder finished +15.66% versus DCA. Over 1990–2026 it finished –3.40%. In 1990–1999 it finished –10.28%. Contribution-level omniscience—wait until the next true 10% trough before deploying that month’s dollar—has mean $\Delta = +4.69\%$ and $t = 0.68$. The mean is positive because some months are followed by deep, recoverable crashes. The t -statistic is a coin flip because other months wait through a bull, then buy a local trough that is still a much higher total-return price (mean ΔP while waiting +22%, the same order as the tradable 10% rule). Even God does not get cheap shares in level terms. God gets the local bottom. Over a 36-year drift, that is not enough to pay for the waits that do not precede a lost decade.

The practical content of Exception 6 is a bound. If perfect trough identification of reserved cash cannot beat DCA over the full sample, imperfect RSI, moving-average, and 10% rules should not be expected to. The bound is not “timing can never work in a bear.” It is “the skill required to make reserved cash win unconditionally is more than perfect local bottoms, and in the one decade where even noisy crash rules won, you did not need an oracle to want cash—you needed the decade.”

8.7 Exception 7: “buy the dip” as keep buying, which is DCA

A large part of popular talk uses the same three words for a policy this paper never finds wanting. The policy is: you are already investing; the market falls; you do not freeze, you do not sell, you keep putting the paycheck in. March 2009 and March 2020 then look, in hindsight, like brilliant dip purchases. They are the DCA path. The baseline in every table already includes them. The tests that lost are the other policy: cash is held on purpose until a trigger. The two policies are both called “buying the dip.” Only one is a timing overlay.

This exception is listed last among the “when it works” cases because it is the case that always worked in the paycheck problem, and it is not an exception to the headline—it *is* the headline. It is included so that the taxonomy is complete relative to ordinary language. If an investor’s “buy the dip” plan is an automatic monthly purchase that continues in 2008 and 2020, this paper has no quarrel with the plan. If the

plan is to skip purchases until -20% , the plan is Exceptions 1 and 6 on a good path and Sections 5–6 on the path that actually occurred.

8.8 Joint condition, and three things that are not exceptions

The cases that raise wealth relative to a relevant benchmark can be stacked. Dip-waiting is most helpful when (a) the investor is parked near a rich peak, (b) a long bear or lost decade is coming, and (c) the cash would have sat idle anyway. Condition (a)+(b) is Exception 1 and is not knowable in advance. Condition (c) is Exception 2 and expires the moment the investor would have DCAed. Exception 3 requires abandoning crash definitions. Exception 4 requires already being a scheduled investor. Exception 5 requires a mixed book and rebalancing, not a bill reserve. Exception 6 requires the future. Exception 7 is DCA.

Three comparisons that are often offered as exceptions are not.

A pile waiting for -10% or -20% . Section 6 is the test that matches an investor who has the money today and no future paychecks to dollar-average. Lump sum wins on the median and the mean. Waiting for a dip wins more often than 12-month DCA and loses more when it loses. That is not “buy the dip works for lumps.” It is “buy the dip is a worse insurance policy than spreading the pile, and spreading the pile is already costly insurance versus investing now.”

Short-horizon spending needs. If the cash is for rent, tuition, or a house deposit inside a few years, the S&P 500 is the wrong asset, not the wrong entry rule. In this sample a lump into VFINX was underwater 17% of one-year starts (worst -46%), 15% of three-year starts (worst -43%), and 0% of twenty-year starts (worst still $+169\%$). Lump versus dip does not repair a horizon mismatch. An emergency buffer in bills is Exception 2 applied to money that *should* stay in cash. It is not a timing stash for the long-horizon remainder.

Conglomerate cash and famous investors. A firm that holds T-bills as operating oxygen and as optionality to buy entire businesses at a discount is not running a VFINX 10% overlay. The mandate, the opportunity set, and the outside option differ. Copying the headline “cash on the balance sheet” as “I will wait for the index to drop” is Exception 5 and Exception 7 misapplied: a true idea about not overpaying for a specific cash-flow stream, sold as underinvesting a market portfolio the household can buy today in any size. The household instruction that matches this paper’s baseline is the fully invested index saver, not the unused deal pipeline.

8.9 What an investor is allowed to conclude

After the exceptions, the supported conclusions are narrow and usable.

1. If cash arrives over time and will be invested in a broad U.S. equity index, investing it as it arrives is the robust simple rule. Crash reserves, RSI, moving averages, and wait-for- -20% overlays reduced terminal wealth in this sample, except when they were DCA in disguise or a few tenths of a percent of pull-forward.
2. If a lost decade after a peak is the realized path, deep-crash waiting can win. That path is not identified in advance by the rules in the catalog. Using it as a plan requires believing the next

decade is 2000–2009 rather than 1990–1999 or 2010–2019.

3. If the cash would never have entered equities, buying a dip is much better than bills and still worse than a monthly schedule once the investor is willing to have a schedule.
4. If the cash is already in hand for the long run, the ranking is lump sum, then a short DCA of the pile as sleep insurance, then wait-for-a-dip last on typical outcomes.
5. If “buy the dip” means “do not stop the schedule in a crash,” the paper agrees. That meaning should not be sold as a cash reserve.

None of these items is a license to treat the exceptions as a menu from which a household picks the one that feels like skill. The only exception that is free, always available, and not a forecast is Exception 7: keep buying. The only exception that is a small same-cash timing gain is Exception 4, and it is economically empty. The only exception that looks like the influencer chart is Exception 1, and it requires the decade that hurts to live through and that one cannot condition on without also conditioning on the decades that make crash-waiting ruinous.

9. Conclusion

Buy-the-dip as a delay of S&P 500 contributions is not a free overlay on dollar-cost averaging. In a 36-year daily sample with a fair-capital constraint, it subtracts terminal wealth in proportion to how long cash sits and how rare the trigger is, because delayed paychecks are filled at later total-return prices, not cheaper ones. The pile version of the same idea loses to lump sum. Those rankings are sample facts, not a proof about the next decades: they rest on a realized U.S. equity premium, and a different premium would change the arithmetic. The exceptions are real and they are narrow. They occupy Section 8 so that they are not footnoted out of existence and so that they are not allowed to swallow the result. For a long-horizon saver who will keep putting money into the index, the policy that held up in this record is the one that never waited.

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Data availability

Daily prices are from Yahoo Finance: VFINX total return, the S&P 500 price index, and the 13-week Treasury bill. Programs that construct the forward total-return series, run the strategy catalog, and produce the contribution-level tests and figures are in the accompanying replication folder.